



Kenya Certificate of Secondary Education (KCSE)

441/1
HOME SCIENCE
PAPER 1
Theory
TIME: 2 ½ HOURS

MARKING SCHEME

SECTION A-(40MKS)

1. **Qualities in hand sewing needle.** (2mks)
 - Fairy sharp and fine to easily slope through the fabric without leaving behind ugly marks.
 - Stainless steel free from rust which would otherwise stain and discolour the fabric.
 - Smooth eye for easy threading, rough eye easily cuts the thread.
 - Correct size in relationship to the purpose it is purchased for.

(2x1=2mks)
2. **Disadvantages of relying on solar energy in the home.** (3mks)
 - The battery requires frequent replacement and this can be expensive.
 - In case of prolonged cloud cover, the means shortage of solar power.
 - Its use is mainly limited to lightening and operating of electricity appliances but not cooking.

(1x3=3mks)
3. **Two conspicuous seams.** (2mks)
 - Double stitched seam.
 - Overlaid seam.

(1x2=2mks)
4. **Qualities of a good dust bin.** (2mks)
 - Should be strong
 - Well fitting lid.
 - Large enough.
 - Rust proof.

(Any two 2x1=2mks)
5. **Coating food before deep frying** (3mks)
 - Prevent from breaking into pieces.
 - Prevent from overcooking.
 - Improve its appearance.
 - Prevent food from being too greasy.

(Any 3x1=3mks)
6. **Causes of anaemia.** (3mks)
 - Genetic (sickle cell)
 - Excessive loss of blood.
 - Sickness e.g. malaria leukaemia.
 - Parasites e.g. hook worm

(Any 3x1=3mks)
7. **Uses of fats in cookery.** (3mks)
 - Cut short gluten strands.
 - Soften mixtures.
 - Add flavour and colour.
 - Increase shelf life.

(Any 3x1=3mks)



8. **Reasons for blending vegetables before freezing.** (3mks)
- Help retain colours.
 - Reduce bulkiness.
 - Retain nutrients.
 - Discontinue the working of enzymes.
- (Any 3x1=3mks)
9. **Two disadvantages of convenience food.** (2mks)
- A lot of additives e.g Preservatives, colouring.
 - More expensive than French foods.
 - Nutritive value maybe inferior to that of French food.
 - Can easily cause food poisoning if poorly preserved.
 - Shelf life has expired.
 - Likely to be contaminated during processing and packaging.
- (Any 2x1=2mks)
10. **Two reasons why a lactating mother would have inadequate milk.** (2mks)
- Due to psychological seam.
 - Poor diet e.g. less fluids
 - Sickness.
 - Mother not breast feeding frequently.
- (Any 2x1=2mks)
11. **Why clothes are steeped.** (2mks)
- Loosen dirt.
 - Dissolved water based stains.
 - Soften starch.
 - Wetting the fabric to make working easier.
- (1x2=2mks)
12. **Why interfacing is done on collars, cuffs and bands .** (2mks)
- Bring out edges of the area.
 - Gives them body.
 - Prevents sagging of the area
- (Any 2x1=2mks)
13. **Reasons why a consumer needs protection.** (4mks)
- Goods and services are of good quality.
 - Goods and services are provided at the correct quantities.
 - Prices charged for goods and services are fair .
 - The consumer gets right information.
 - Health of the consumer is maintained.
 - Commodities are available.
 - The consumer enjoys the right to compensation.
- (any 4x1=4mks)
14. **Four types of weaning.** (2mks)
- Abrupt weaning.
 - Partial weaning.
 - Gradual weaning.
 - Natural weaning.
15. **Dangers of excess weight gain during pregnancy.** (3mks)
- Can cause varicose veins.
 - Can cause stretch marks.



- Increase fatigue in an expectant mother.
- Increase pressure on the heart.
- One may not loose weight after delivery.

(Any 3x1=3mks)

16. Desirable qualities of baking flour.

(2mks)

- Should have correct amount of gluten.
- Finely ground.
- No foreign bodies.
- Correct colour.
- Without lumps.
- Fresh with no odour of staleness/not expired.

(Any 4x ½ =2.)

17. Why a shopping list is useful to the consumer

(1mk)

- To ensure that no item is forgotten.
- Helps one to keep within the list items.
- Reduces chances of overspending.

(Any 2x ½ =1mk)

Section B-20mks (Compulsory)

18. a) Launder a white tea towel with an old tea stain.

(8mks)

- i) Cover with borax ½ then pour boiling ½ water through the stain to loosen the stain ½ .
- ii) Soak ½ in cold water ½ to loosen dirt ½
- iii) Wash in hand hot ½ soapy water using friction ½ to remove fixed dirt.
- iv) Rinse severally in warm water ½ to remove traces of soap and dirt ½
- v) Boiling ½ to disinfect.
- vi) Rinse finally ½ in cold water ½ using out excess water.
- vii) Dry in direct sunshine ½
- viii) When dry, press ½ with a fairly hot iron ½

(½ x 16=8mks)

b) Wash a neglected aluminium pan

(4mks)

-Wash in hot ½ soapy ½ water using scouring pad ½ to remove stains ½ rinse ½ severally ½ using hot ½ water. Wipe dry ½ with a kitchen cloth to prevent water mark.

(½ x 8 =4mks)

c) Weekly clean the kitchen floor made of terrazzo.

(8mks)

- i) Sweep ½ the floor to remove loose dirt ½ dust the skirting board ½ .
- ii) Scrub ½ using warm ½ soapy ½ water working a small section ½ at a time.
Overlapping ½ the section to ensure complete removal of dirt ½
- iii) Clean with a cloth ½ wrung out of warm 12 water to remove traces of soap and dirt ½
- iv) Dry thoroughly ½ with a dry floor cloth ½
- v) Rub hard ½ with a dry cloth to buff ½



Section C 40 mks.

19. a) Precautionary measures to take while deep frying foods. (5mks)

- Do not fill the pan more than $\frac{2}{3}$ full to avoid oil overflow.
- Lower the food gently into the oil to avoid splashing on the ground.
- Heat the fat Just to the right temperature to avoid catching fire.
- Avoid spilling water into the hot fat to prevent fire outbreak/burning.
- Balance the pan / sufuria well on the cooker to prevent accidents as it may tipple over.
- Keep the fat away from reach to prevent burning for it to cool.

(1x5=5mks)

b) Ways of finishing the lower edge of a short sleeved blouse decoratively. (5mks)

- Scalloping /shell edging.
- Binding with contrasting colour
- Facing with contrasting colour.
- Putting a frill.
- Applying lace.
- Embroidery of loop stitchery
- Crocheting.

c) Five points a consumer should consider before buying an item (5mks)

- The cost of the item should be affordable.
- The quality of the item and its durability
- Buy an item which is versatile. It will be more useful i.e. a kitchen knife should be used for other jobs.
- Style and design of the item in relation to ease and comfort in use.
- The item being bought should be attractive in appearance.
- Window shop to compare prices and qualities.
- Do not purchase on impulse.
- Read labels.

(1x5=5mks)

d) Considerations to make when finishing a living / sitting room. (5mks)

- Money available
- Furnishing should be easy to care for.
- Furnishing should conform to the colour scheme.
- Furnishing should be of good workmanship
- Size of the furnishing should be suitable to the size of the room.
- The furnishings should have attractive design.
- The furnishing should be durable.

(5x1=5mks)



20. a) Explain factors that influence the cleaning frequency of a home.

(10mks)

- Amount of dust and dirt brought in it this can vary according to its location and season 2mks
- Use of the house or particular room in the house e.g toilets, bedrooms, kitchen need more frequent attraction than other rooms.
- The occupants e.g. a house with young children needs more attention because they make it dirty & hygiene should be at it highed.
- The conditions of the home or surface e.g. cracked floors need frequent cleaning than smooth ones they collect more dust than smooth ones.
- Available time for cleaning.
- Cleaning equipments and material available.

(10mks)

b) Points to observe when fixing fastening onto a garment.

(5mks)

- Fasteners must be shown on double fabric.
- Be sufficient in number so that opening can be kept closed.
- Be evenly spaced e.g. buttons.
- Be sown directly opposite each other so that the opening will be flat when fastened.
- For thick fabrics, sufficient space be allowed between button and overlap.

1x5=5mks

c) Ensuring that left over foods do not go bad in the absence of a refrigerators. (5mks)

- Place in small containers.
- Should be well covered.
- Cool before placing in a cupboard.
- Wrap in grease proof paper, foil and polythene.
- Discard any food showing figures of spoilage.

21. a) Making a single pointed dart.

(5mks)

- Mark the position of the dart.
- Fold the fabric R.S together on the W.S and tack.
- Machine stitch along the tacking beginning from the wide part of the tapering end.
- Remove tacking stitches.
- Press the dart to the centre back & the centre front.

(5mks)

b) Signs and symptoms of pregnancy.

(5mks)

- Stopping menstrual periods.
- Nausea or morning sickness during the first 3-4 months of pregnancy.
- Frequent desire to pass urine .
- Strong like or dislike of certain foods
- Enlargements of breasts.



- Tenderness of nipples.
- Bulging the abdomen.
- Dark line on the abdomen running from the navel downwards.
- Movement of the foetus which is felt at about fifth month of pregnancy.
- Some changes in skin complexion. The mother may become smooth and lighten or the opposite.
- Some expectant mothers experience heart burn or frequent constipation

c) Food hygiene practice.

- Wear protective clothing to prevent contaminating from out
- Cover the hair to prevent it falling in food.
- Wash hands well before handling food to avoid contaminating food.
- Avoid habits like testing food using fingers to avoid dirt getting into the mouth
- Persons suffering from diarrhoea, typhoid, cholera, dysentery, T.B e.t.c should not handle food.
- Kitchen equipments should be clean to avoid dirt getting into the food.

(10mks)